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Scenario

PART 1: Animalia Animal Shelter Database System

Animalia Animal Shelter is a non-profit organization located in Sunway City, Malaysia. Founded in 2020, it was established with the aim of rescuing, rehabilitating, and rehoming stray, surrendered, and injured animals. The shelter houses mainly cats and dogs but occasionally admits rabbits and other small mammals. The shelter operates with a small administrative staff, several volunteers, and veterinary partners. Although the organization receives partial funding through local government grants, most of its expenses are covered through public donations.

Currently, all operational records are maintained manually using excel spreadsheets and logbooks which has led to various issues such as delays, data loss, difficulty tracking medical histories and a lack of real-time visibility into animal status. These problems have made it difficult to coordinate adoption and foster in a structured process, track donation history, and assign volunteers efficiently.

To overcome these challenges, Animalia has decided to implement a relational database system to centralize all information and make it accessible and maintainable. The primary objectives of this new database system are to facilitate tracking of health records, streamline adoptions and fosters, record donations as well as manage volunteer assignments in an integrated way.

The main operations managed by the Database are:

1. Animal Intake and Records

Each animal is registered into the system with a unique ID and their basic information is recorded (e.g. species, breed, gender, age, arrival date). It also contains the status of the animal which can be either “Available”, “Fostered”, “Adopted”, or “Under Treatment”.

2. Health Checkups

Each animal must go through at least one health check which is carried out by a trained staff or veterinarian. The health record includes date, condition description, rating (1-5), and the staff responsible for the medical checkup.

3. Foster and Adoption Programs

Many animals are placed in temporary foster homes before adoption to recover from injuries, adapt to home environments, or free up space in the shelter. The system tracks these foster arrangements by storing the start and end dates, caregiver information, and

medical notes in the foster record. Foster caregivers are registered under the caregiver entity.

When an animal is ready for adoption, interested individuals also register as Caregivers and go through a vetting process. If approved, the adoption is recorded with the caregiver's ID, animal ID and adoption date.

A caregiver who fosters an animal may later adopt the same animal, provided they complete the official adoption process.

4. Donations and Donors

Donor details are registered with unique IDs and may contribute multiple times. Donations are recorded with details such as donor ID, amount, Purpose (e.g. food, vet services) and date.

5. Staff Management

Staff are responsible for maintaining records such as animal entries and health checks.

Shelter staff includes administrators, caretakers, and veterinary assistants. Each staff member has a profile in the system with their ID, name, position, contact, and join date.

6. Volunteer Involvement

Animalia depends on volunteers to support their daily activities and proper animal care.

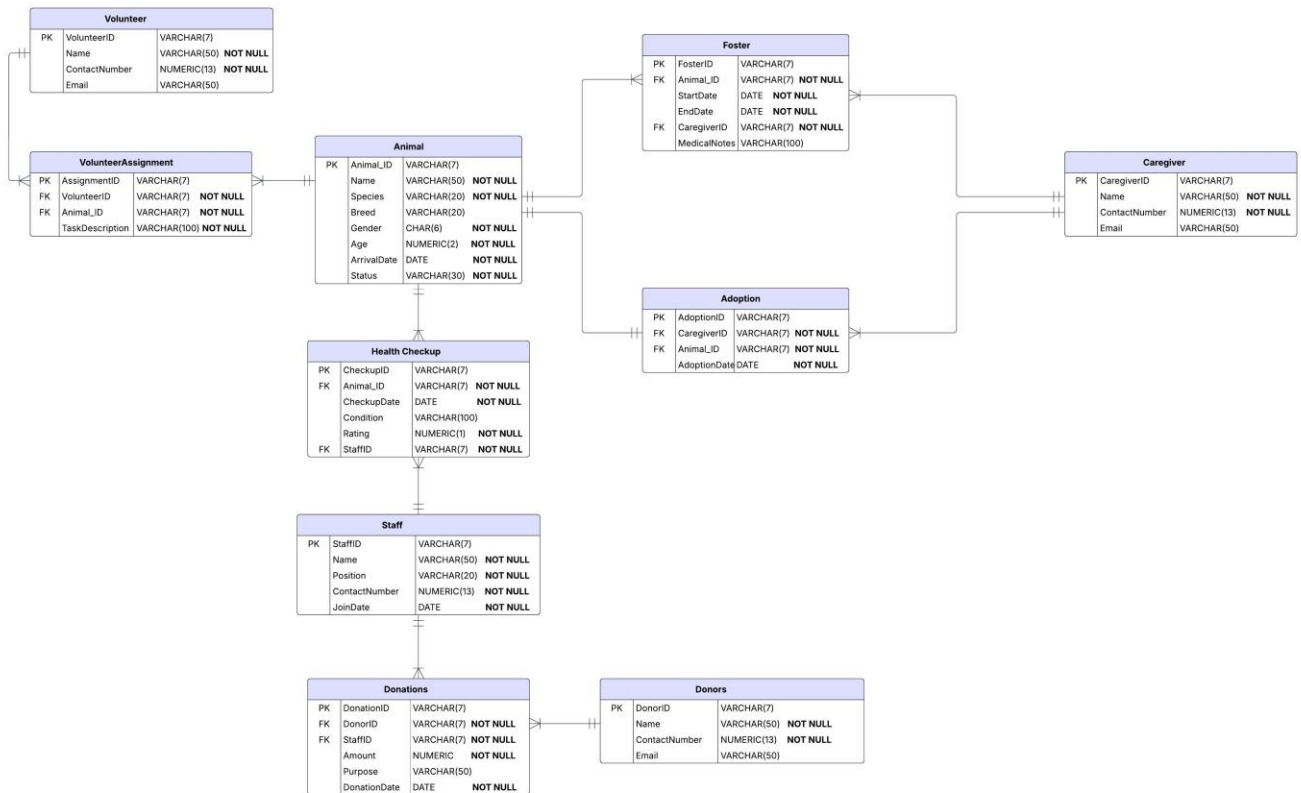
Volunteers can be assigned to assist with feeding, cleaning, playtime, or general care.

Each volunteer is registered in the system with unique ID and their personal information. To manage this effectively, the system uses a Volunteer_assignment feature that records which volunteer is currently in charge of which animal. This allows the shelter to track the participation of volunteers.

Business Rules

1. An animal can only be adopted if its status is marked as available and has undergone at least 1 health check.
2. A volunteer can be assigned to multiple animals, and each animal can have multiple volunteers
3. One caregiver can take responsibility for multiple animals, but each animal can only be adopted once and fostered by one caregiver at a time.
4. All health check records must be entered into the system by a registered member of staff.
5. Each donation record must be handled and processed by a staff member.
6. One animal can have multiple health check records, but each health check must be linked to one animal and one staff member.
7. Only animals marked with available status can be assigned to volunteers.
8. Every Donation record must be associated with a registered staff member who assisted or processed the donation.

Entity Relationship Diagram



Entities: Animal, adoption, foster, caregiver, volunteer, volunteer_assignment, Health Checkups, Staff, Donation, Donor

Junction Tables:

Foster: links animal to caregiver (many: many)

Volunteer_assignment: links volunteers to animal (many: many)

ERD Relationships

Caregiver-Adoption: A caregiver can adopt multiple animals, but each animal can only be adopted once. (many:1)

Animal-adoption: An animal can be adopted only once and each adoption is associated with one animal. (1:1)

Animal-Foster: An animal can have multiple foster records provided the foster periods do not overlap but each foster record is for one animal. (1: many)

Caregiver-foster: A caregiver can foster many animals, and each animal can be fostered many times by different caregivers as long as the foster period of these caregivers do not overlap. An animal can be fostered by only 1 caregiver at a time. (1: many)

Volunteer- VolunteerAssignment-Animal: A volunteer can be assigned to many animals and each animal (animal must be marked as available) can have multiple volunteers. (many: many)

Animal-Health Checkups: An animal can have multiple health checkups but each health checkup is associated to one animal only (1: many)

Health checkup/Donation - staff: Each health checkup/donation record is handled by 1 staff member but 1 staff member can handle many health checkups records/donations. (1: many)

Donor-Donation: A donor can make multiple donations but each donation is linked to one donor only.

Constraints

	Constraint	Justification
1	Animal Status Constraint CHECK (Status IN ('Available', 'Fostered', 'Adopted', 'Under Treatment'))	This constraint only allows values such as: 'Available', 'Fostered', 'Adopted', and 'Under Treatment'. This ensures that only valid status is entered and safeguards any misclassification or data entry errors that might occur when updating the data; i.e, assigning volunteers to already adopted animals.
2	Health Rating Constraint CHECK (Rating BETWEEN 1 AND 5)	This constraint ensures that the ratings stay within the correct range, avoiding misinput of wrong values that are not existent within the database.
3	Donation Amount Constraint CHECK (Amount > 0)	It does not make sense for donors to be able to donate no amount of funds, but also for there to be any negative values. Thus, this prevents donations with zero or negative amounts, so financial records stay accurate for reporting and audits.
4	Foster Period Constraint CHECK (EndDate >= StartDate)	Using the more than equal to function avoids errors like ending a foster before it has even begun.
5	Staff Role Constraint CHECK (Position IN ('Administrator', 'Caretaker', 'Vet Assistant'))	This limits the entries to the known staff roles, keeping the staff list consistent and prevents typos or new and unrecognized roles from being saved within the database.
6	Animal Age Constraint CHECK (Age BETWEEN 0 AND 40)	Ensures that only realistic animal ages are recorded within the database. This prevents data entry errors like negative ages or high values (i.e., 1000 years old).
7	Animal Gender Constraint CHECK (Gender IN ('Male', 'Female'))	Limits the entries to just "Male" or "Female" for animal genders. It avoids errors like typos. For example, typing "female". In addition, this ensures data accuracy within the database records and helps with grouping for medical and/or adoption.